

Mathematical Tools for Physics Syllabus

2025 2nd semester

Title	Introduction to Relativistic Cosmology		Code	PE2105	Grades	3	Classroom	
			Course	Elective	Hours	3		
Instructor	Name	Class	Office	Office hour	Contact	e-mail		
	김현벽		Chang 5306			kimhyunbyuk@ksa.hs.kr		

1. Outline

Course Objective
After completing this course, students should be able to • apply mathematical methods to solve physics problems at the level of general physics 1 & 2. • use vectors to analyze mechanics problems. • use complex variables to solve electromagnetic problems. • utilize Fourier analysis and statistical methods for advanced problems. • solve ordinary and partial differential equations of mechanics and electric circuit problems.
Course Description
■ Complex variables Phasor methods for various physics problems will be introduced. ■ Vector algebra Algebraic calculations and geometric meaning of vectors in Newtonian mechanics will be treated. Elementary differentiation and integration will be also discussed. ■ Vector calculus We will master divergence theorem and Stoke's theorem which are relevant to classical electrodynamics. Laplace and Poisson's equations will be also studied. ■ Fourier analysis Fourier series and Fourier integrals for wave functions will be introduced. ■ Differential equations Boundary and initial value problems associated with mechanics, heat, wave will be studied. ■ Statistical methods Mathematical techniques which are relevant to the counting of the number of states in statistical physics will be studied.

2. Core Target Competencies

Analytical and Creative Thinking	Critical Thinking	Integrity
Basic knowledge is dealt in the class. But the main point is not to deliver the established facts but to train the thinking method. Students are heavily encouraged to sharpen their critical arguments and in-depth discussion skills. Moreover, having a firm big picture of the subject is one of the ultimate goals of the course.		

3. Text

- A. Textbook : Mathematical Methods in the Physical Sciences 3rd (Boas, Wiley)
 B. Reference : Mathematical Methods for Physics and Engineering 3rd (Riley et al., Cambridge)

4. Class Method

- A. Lecture, Discussion & Debate

5. Evaluation (%)

Midterm	Final	Quiz	Homework [☆]	Attendance & Attitude [*]	Total
30	30	10	20	10	100

☆ Penalty will be given to the overdue homework. : -50% after due date

※ Penalty : - 0.3 point per late & - 1.0 point per absence without notice.

※※ 공결 및 병결 등의 사항은 학업성적관리규정을 따름

6. Grade (절대평가)

$A+ \geq 80\%$, $80\% > A0 \geq 75\%$, $75\% > A- \geq 70\%$,
 $70\% > B+ \geq 65\%$, $65\% > B0 \geq 60\%$, $60\% > B- \geq 55\%$,
 $55\% > C+ \geq 50\%$, $50\% > C0 \geq 45\%$, $45\% > C- \geq 40\%$,
 $40\% > D+ \geq 35\%$, $35\% > D0 \geq 30\%$, $30\% > D- \geq 25\%$, $25\% > F$

7. Schedule

Week	Subject	Homework
1	Real analysis	Handout
	Analytical and Creative Thinking, Critical Thinking, Integrity	
2	Real analysis	Handout
	Analytical and Creative Thinking, Critical Thinking, Integrity	
3	Complex variables	Handout
	Analytical and Creative Thinking, Critical Thinking, Integrity	
4	Complex variables	Handout
	Analytical and Creative Thinking, Critical Thinking, Integrity	
5	Vector algebra	Handout
	Analytical and Creative Thinking, Critical Thinking, Integrity	
6	Vector algebra	Handout
	Analytical and Creative Thinking, Critical Thinking, Integrity	
7	Vector calculus	Handout
	Analytical and Creative Thinking, Critical Thinking, Integrity	
8	연휴	
9	Midterm Exam	
10	Vector calculus	Handout
	Analytical and Creative Thinking, Critical Thinking, Integrity	
11	Vector calculus	Handout
	Analytical and Creative Thinking, Critical Thinking, Integrity	
12	Fourier analysis	Handout
	Analytical and Creative Thinking, Critical Thinking, Integrity	
13	Fourier analysis	Handout
	Analytical and Creative Thinking, Critical Thinking, Integrity	
14	Differential equations	Handout
	Analytical and Creative Thinking, Critical Thinking, Integrity	
15	Differential equations	Handout
	Analytical and Creative Thinking, Critical Thinking, Integrity	
16	Statistical methods	보강 주간
	Analytical and Creative Thinking, Critical Thinking, Integrity	
17	Final Exam	

*** Schedules can be changed under circumstances.